

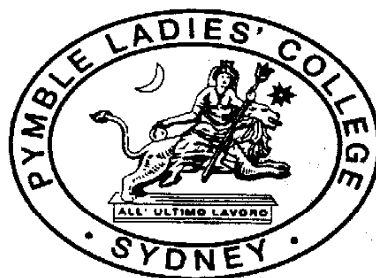
Mrs Gibson
Mr Keanan Brown
Mrs Lee
Mrs Choong
Mrs Leslie

PYMBLE LADIES' COLLEGE

YEAR 12

MATHEMATICS EXTENSION 1

HSC TRIAL EXAMINATION 2001



Time Allowed: 2 hours + 5 mins reading time

Test date: 16 August 2001

Instructions:

- All questions should be attempted.
- Write your name and your teacher's name on each page
- Start each question on a new page.
- **DO NOT** staple the questions together.
- Only approved calculators may be used.
- A standard integral sheet is attached.
- Marks might be deducted for careless or untidy work.
- Hand this question paper in with your answers.
- ALL rough working paper must be attached to the back of the last question.
- Staple a coloured sheet of paper to the back of each question.
- There are seven (7) questions in this paper.
- All questions are of equal value.

MARKING GUIDELINES

- Provide answers which are complete, accurate and comprehensive.
- Leave your answers in exact form unless otherwise stated.
- Include all necessary working. Correct answers will not necessarily gain full marks unless necessary working is shown. Relevant working might gain marks even if your answer is wrong.
- Take care with mathematical notation.
- Show relevant information clearly and unambiguously on sketches if required.
- Present well set out solutions using a logical set of steps in which justification is included where necessary.

QUESTION 1**Marks**

- (a) Differentiate $\frac{1}{1+x^2}$ **1**
- (b) The polynomial $P(x) = 2x^3 - x + a$ is divisible by $x + 2$. **1**
Find the value of a .
- (c) A, B and P are the points $(-1,8)$, $(6,-6)$ and $(4,-2)$ respectively. **2**
The point P divides the interval AB internally in the ratio $k:1$.
Find the value of k .
- (d) Solve $x - 1 = \sqrt{x + 1}$ **3**
- (e) Evaluate $\int_1^{\sqrt{2}} \frac{1}{\sqrt{4-x^2}} dx$ **3**
- (f) Solve $|3 - 3x| > x + 3$ **2**

QUESTION 2**Start a new page****Marks**

(a) Find the exact value of $\cos\left(\sin^{-1}\left(-\frac{1}{3}\right)\right)$ **2**

(b) Given that $\log_b a = 2$ and $\log_c b = 3$, find the value of $\log_a c$. **2**

(c) Find the value of $\int_0^3 \frac{t}{\sqrt{1+t}} dt$ **4**

using the substitution $t = u^2 - 1$ where $u > 0$

(d) A and B are acute angles such that $\cos A = \frac{3}{5}$ and $\sin B = \frac{1}{\sqrt{5}}$. **4**

Without finding the size of either angle, show that $A = 2B$, and

use this result to find the exact value of $\sin 3B$.

QUESTION 3**Start a new page****Marks**

- (a) Write down the value of the **prime** number b such that

$$\sum_{n=1}^3 \log_2 2n = a + \log_2 b$$

1

- (b) The diagram shows two circles touching externally at T .

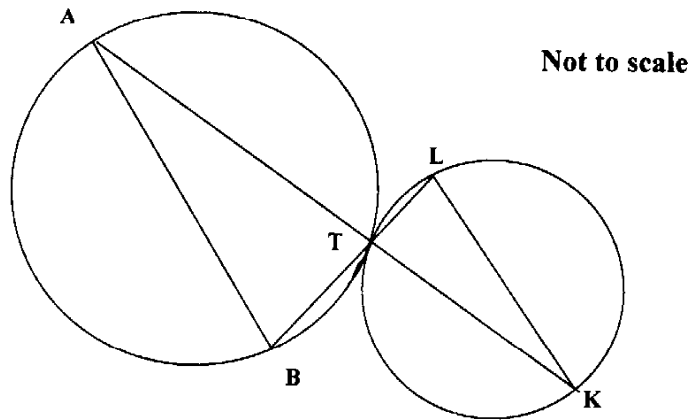
AB is any diameter of the first circle, and AT and BT are produced to meet the second circle again at K and L respectively.

Copy the diagram onto your answer paper, then prove that

- (i) KL is a diameter of the second circle

1

- (ii) LK is parallel to AB

2

- (c) Evaluate $\int_0^{\frac{\pi}{4}} (\cos x + \sec x)^2 dx$

4

- (d) The perimeter of an equilateral triangle of side a cm is increasing at a constant rate of 6 cm/sec as the triangle is being enlarged.

4

Find the rate at which the area of the triangle is increasing at the instant the perimeter is 24 cm. (The triangle remains equilateral.)

QUESTION 4**Start a new page****Marks**

- (a) A certain population N is changing at a rate given by the equation

$$\frac{dN}{dt} = 0.5 (N - 100).$$

- (i) Show that $N = 100 + Ae^{0.5t}$ is a solution of this equation, and find the value of A given that the initial value of N is 500. **2**
- (ii) Find the value of N when $t = 10$. **1**

- (b) A function $f(x)$ has an inverse whose equation is $f^{-1}(x) = \frac{2x-2}{x-2}$. **3**

What is the equation of $f(x)$?

Explain the geometrical significance of your answer.

- (c) (i) Sketch $f(x) = \sin x$ and its inverse $g(x) = \sin^{-1} x$ on the same axes for $0 \leq x \leq \frac{\pi}{2}$. **1**
- (ii) Show that the tangent at $x = 1$ on $f(x)$ and the tangent at $y = 1$ on $g(x)$ are equally inclined to $y = x$. **4**
- (iii) What is the angle between these two tangents? **1**

QUESTION 5**Start a new page****Marks**

- (a) A particle travels in a straight line executing simple harmonic motion about O according to the equation $x = a \cos nt$.
- (i) Show that the velocity v and displacement x of the particle at any time t are related by the equation $v^2 = n^2(a^2 - x^2)$. 2
- (ii) Hence show that the acceleration of the particle can be given as $\ddot{x} = -n^2x$. 1
- (b) A particle executes simple harmonic motion about O according to the above equations. Initially it is at $x = 2$. As it passes through O its speed is 2 m/sec. How long does it take to get to O for the first time? 3
- (c) Draw a large and accurate sketch of the curve $y = \frac{x+4}{x(x+8)}$, showing all essential features such as intercepts on axes and asymptotes. 4
- Show that there are no stationary points. (You do not need to find the coordinates of any inflection points.)
- (d) Find the area bound by the curve $y = \frac{x+4}{x(x+8)}$ and the x -axis between $x=1$ and $x=2$. 2

You may use the substitution $u = x(x+8)$ to evaluate this area if you wish.

QUESTION 6**Start a new page****Marks**

(a) A curve has equation $f(x) = 3x - 4x^3$.

(i) Show that the equation of a tangent at the point on the curve

2

where $x = a$ is $y = (3 - 12a^2)x + 8a^3$.

(ii) How many tangents can be drawn to this curve from the point (1,0)?

3

(You must show full working to substantiate your answer.)

(b) P $(2ap, ap^2)$ and Q $(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$.

The tangent at P and a line through Q parallel to the y axis meet at point R.

The tangent at Q and a line through P parallel to the y axis meet at point S.

(i) Draw a neat diagram showing all information given above.

1

(ii) Show that the equation of the tangent at P is $y = px - ap^2$.

2

(iii) Show that PQRS is a parallelogram

2

(iv) Show that the area of PQRS is $2a^2|p - q|^3$ square units.

2

QUESTION 7**Start a new page****Marks**

- (a) A particle moves in a straight line towards the centre O experiencing an acceleration that is inversely proportional to the cube of the distance from O,

namely $a = -\frac{4}{x^3}$.

- (i) If the particle starts from rest at $x=2$, find an expression for the velocity of the particle in terms of x . 3

Make sure you justify the sign of your expression.

- (ii) Hence find an expression that relates elapsed time t and displacement x , and find the time the particle takes to reach $x=1$ (for the first time, if it does so more than once). 3

- (b) (i) Prove by induction that for all integers $n \geq 1$ 3

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

- (ii) Use this result to evaluate $2^2 + 4^2 + 6^2 + \dots + 100^2$ 2

- (iii) Hence evaluate $1^2 + 3^2 + 5^2 + \dots + 99^2$ 1

END OF PAPER

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

(21) (a) $\frac{d}{dx} \left(\frac{1}{1+x^2} \right) = \frac{-2x}{(1+x^2)^2}$ (1)

11

(b) $P(x) = 2x^3 - x + a$
 $P(-2) = 0 \Rightarrow 2(-2)^3 - (-2) + a = 0$
 $-16 + 2 + a = 0$
 $\therefore a = 14$ (1)

11

(c) $\begin{matrix} A & P & B \\ x: & -1 & 4 & 6 \end{matrix}$

12

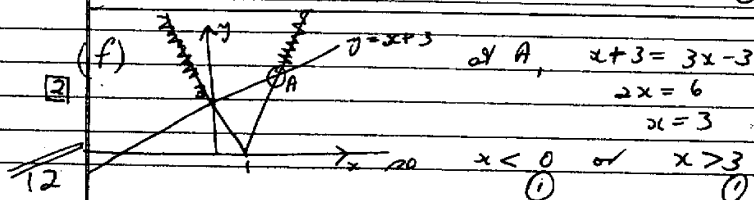
$\frac{AP}{PB} = \frac{5}{2} = \frac{2\frac{1}{2}}{1} \therefore k = 2\frac{1}{2}$ (2)

(d) $x-1 = \sqrt{x+1}$ $x-1 \geq 0$
 $x \geq 1$
 $(x-1)^2 = x+1$
 $x^2 - 2x + 1 = x+1$ (1)
 $x^2 - 3x = 0$
 $x(x-3) = 0$ (1)
 $x = 0$ or 3
 but since $x \geq 1$, only $x = 3$ (1)

3

(e) $\int_1^{\sqrt{2}} \frac{1}{\sqrt{4-x^2}} dx = \sin^{-1} \frac{x}{2} \Big|_1^{\sqrt{2}}$ (1)
 $= \sin^{-1} \frac{1}{\sqrt{2}} - \sin^{-1} \frac{1}{2}$
 $= \frac{\pi}{4} - \frac{\pi}{6}$ (1)
 $= \frac{\pi}{12}$ (1)

3



3

(22) (a) $\cos^{-1}(-\frac{1}{3}) = \cos \alpha$
 $= \frac{2\sqrt{2}}{3}$ (2)

2

$\frac{1}{3}$ quadrant
 $\frac{1}{3}$ sign
 $\frac{1}{3} \sqrt{8} = \frac{2\sqrt{2}}{3}$
 $\frac{1}{3}$ ratio

(b) $\log_b a = 2 \Rightarrow b^2 = a$ (1)
 $\log_c b = 3 \Rightarrow c^3 = b$ (1)
 $\therefore c^6 = b^2 = a \Rightarrow c = a^{\frac{1}{6}}$ (2)

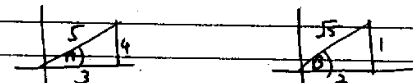
2

(c) $\int_0^3 \frac{t}{\sqrt{1+t}} dt$
 $= \int_1^2 \frac{(u^2-1) 2u du}{\sqrt{u^2}}$ (1)
 $= 2 \int_1^2 (u^2-1) du$ (1)
 $= 2 \left[\frac{u^3}{3} - u \right]_1^2$ (1)
 $= 2 \left[\frac{8}{3} - 2 - \frac{1}{3} + 1 \right]$
 $= 2 \left(\frac{4}{3} \right)$ (1)
 $= \frac{8}{3}$

4

$t = u^2 - 1$
 $dt = 2u du$
 $t = 0, u = 1$
 $t = 3, u = 2$

(d) $\cos A = \frac{4}{5}$ $\sin B = \frac{3}{5}$



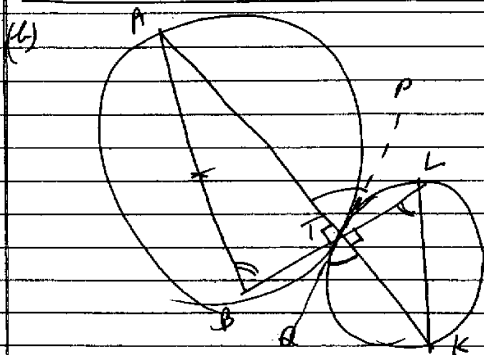
14

$\cos 2B = 1 - 2\sin^2 B$
 $= 1 - 2\left(\frac{3}{5}\right)^2$
 $= \frac{1}{5} = \cos A \therefore A = 2B$ (1)
 $\sin 3B = \sin(B+2B) = \sin(B+A)$
 $= \sin B \cos A + \cos B \sin A$ (3)
 $= \frac{3}{5} \cdot \frac{4}{5} + \frac{4}{5} \cdot \frac{3}{5} = \frac{11}{5}$

12

Q3 (a) $\sum_{n=1}^3 \log_2 2n = \log_2 2 + \log_2 4 + \log_2 6$
 $= 1 + 2 + \log_2 6$
 $= 1 + 2 + (\log_2 2 + \log_2 3)$
 $= 4 + \log_2 3$

$\therefore L = 3$ (1)

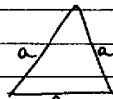


(1) Since AB is diameter
 $\angle ATB = 90^\circ$ (L in semi-circle)
 $\therefore \angle LTK = 90^\circ$ (vert opp $\angle$'s)
 Hence $\angle LTK$ is also $\angle$ in a semi-circle at circum.
 so LK is diameter (1)

(2) Construct tangent PTQ
 $\angle PTA = \angle TBA$ (L between tangent & chord = L in alt. seg)
 similarly $\angle QTK = \angle FLK$ (---) (1)
 but $\angle PTA = \angle QTK$ (vert. opp)
 $\therefore \angle TBA = \angle TLK$ (1)
 these are alternate equal $\angle$'s
 $\therefore AB \parallel LK$

(c) $\int_0^{\frac{\pi}{2}} (\cos x + \sec x)^2 dx$ $\cos 2x = 2\cos^2 x - 1$
 $= \int_0^{\frac{\pi}{2}} (\cos^2 x + 2 + \sec^2 x) dx$ (2)
 $= \int_0^{\frac{\pi}{2}} \frac{1}{2}(1 + \cos 2x) + 2 + \sec^2 x dx$ (2)

(4) $= \left[\frac{1}{4} \sin 2x + \frac{5x}{2} + \tan x \right]_0^{\frac{\pi}{2}}$ (2)
 $= \left(\frac{1}{4} + \frac{5\pi}{8} + 1 \right) - 0$ (1)
 $= \frac{5}{4} + \frac{5\pi}{8}$

(d)  $P = 3a$
 $\frac{dP}{dt} = 3 \frac{da}{dt} = 6 \therefore \frac{da}{dt} = 2$

$A = \frac{1}{2} a^2 \sin 60$
 $= \frac{\sqrt{3}}{4} a^2$ (1)
 $\frac{dA}{dt} = \frac{dA}{da} \cdot \frac{da}{dt}$ (1)
 $= \frac{\sqrt{3}}{2} a \cdot 2$ (1)
 $= \sqrt{3} a$
 $= 8\sqrt{3} \text{ cm} \quad P = 24$ (1)

12

24) (a) $\frac{dN}{dt} = 0.5(N-100)$

(i) $N = 100 + Ae^{0.5t} \Rightarrow Ae^{0.5t} = N - 100$
 $\frac{dN}{dt} = 0.5Ae^{0.5t}$ ①
 $= 0.5(N-100)$ as required

when $t=0$, $N=500$

3) $\therefore 500 = 100 + Ae^0$
 $\text{so } A = 400$ ①

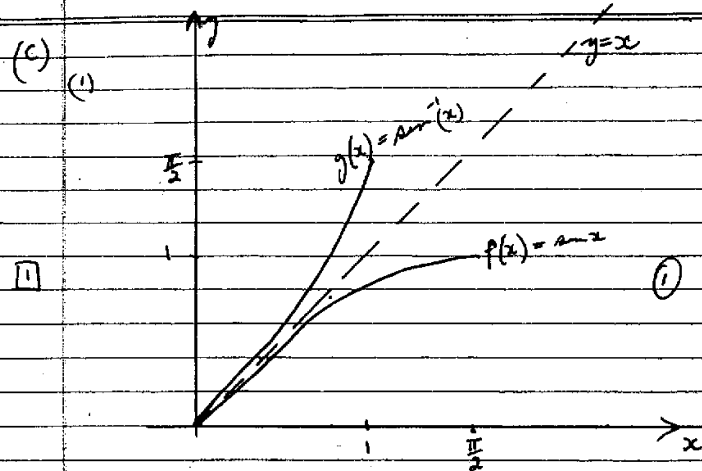
(ii) $N = 100 + 400e^{0.5t}$
 when $t=10$,
 $N = 100 + 400e^5$
 $= 59465.26 \dots$ ①

A) $f^{-1}(x) = \frac{2x-2}{x-2} = y$
 so $f(x)$: $x = \frac{2y-2}{y-2}$ ①

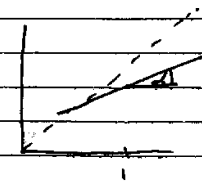
$xy - 2x = 2y - 2$
 $y(x-2) = 2x-2$
 $y = \frac{2x-2}{x-2} = f(x)$ ①

Since $f(x) = f^{-1}(x)$ then the function is symmetric about $y=x$ ①
 as the fn is the inverse of itself

0



(ii) $f(x) = \sin x$
 $f'(x) = \cos x$
 $= \cos 1$ at $x=1$



$\therefore \tan \alpha = \cos 1$
 $\alpha = 0.49536 \dots$ ①

4) so angle between tangent & $y=x$ is $\frac{\pi}{4} - 0.49 \dots$
 $= 0.29^\circ$ ②

gradient of tangent at $y=1$ on $g(x)$ is $\frac{1}{\cos 1}$ ①

so $\tan \beta = \frac{1}{\cos 1}$
 $\beta = 1.0754 \dots$ ②
 $\therefore$ angle between tangent & $y=x$ is
 $1.0754 - \frac{\pi}{4} = 0.29^\circ$ ③

11 (iii) Required angle $= 0.29 \times 2 = 0.58^\circ$ ①

12

(c) Alternative: for (ii)

$$f(x) = \sin x$$

$$f'(x) = \cos x$$

$$= \cos 1 \text{ when } x=1$$

$$= m_1$$

$$\text{So } \tan \alpha = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|$$

$$= \left| \frac{1 - \cos 1}{1 + \cos 1} \right| \quad (1)$$

$$\text{for } g(x) = \sin^{-1} x$$

$$g(x) = y = 1$$

$$g'(x) = \frac{1}{\sqrt{1-x^2}}$$

$$x = \sin 1$$

$$= \frac{1}{\sqrt{1-\sin^2 1}} \text{ at } y=1$$

$$= \frac{1}{\cos 1} \text{ since } 1 - \sin^2 1 > 0 \quad (1)$$

$$= m_2$$

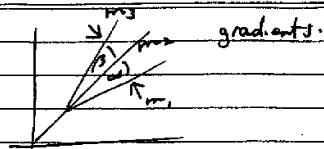
$$\text{So } \tan \beta = \left| \frac{\frac{1}{\cos 1} - 1}{1 + \frac{1}{\cos 1}} \right|$$

(1)

$$= \left| \frac{1 - \cos 1}{\cos 1 + 1} \right| = \tan \alpha$$

$$\therefore \alpha = \beta \text{ as required.} \quad (2)$$

Since α & β are both acute



0

(A5) (a) (i) $x = a \cos nt$

$$v = \frac{dx}{dt} = -an \sin nt$$

$$v^2 = a^2 n^2 \sin^2 nt$$

$$= a^2 n^2 [1 - \cos^2 nt] \quad (1)$$

$$= a^2 n^2 \left[1 - \frac{x^2}{a^2} \right]$$

$$= a^2 n^2 - n^2 x^2 \quad (1)$$

[3] so $v^2 = n^2 (a^2 - x^2)$

(ii) $\ddot{x} = \frac{d}{dt} \left(\frac{1}{2} v^2 \right)$

$$= \frac{d}{dt} \left[\frac{n^2}{2} (a^2 - x^2) \right]$$

$$= \frac{n^2}{2} \cdot -2x$$

$$\ddot{x} = -n^2 x \quad (1)$$

(b) $x = a \cos nt$

$$\text{when } t=0, x=a \therefore a = a \cos 0$$

$$\therefore a = a$$

(1)

$$\text{so } x = a \cos nt$$

$$\text{when } x=0, v^2=4$$

$$\therefore 4 = n^2 (a^2 - 0)$$

$$\therefore n^2 = 1$$

(1)

$$\text{So period} = \frac{2\pi}{n} = 2\pi$$

$$\text{so takes } \frac{1}{2}(2\pi) \text{ secs to get to } 0 \quad (1)$$

$$\text{ie } \frac{\pi}{2} \text{ secs.}$$

OR $x = 2 \cos t$

$$\text{when } x=0, \cos t=0$$

$$\therefore t = \frac{\pi}{2}$$

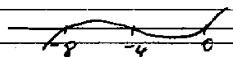
$$(c) \quad y = \frac{x+4}{x(x+8)}$$

when $y=0$, $x=-4$
 $x \neq 0$

vert. asympt. $x=0$ & $x=-8$

horiz. asympt. $y=0$

Sign:



$$\frac{dy}{dx} = \frac{(x^2+8x) - (x+4)(2x+8)}{x^2(x+8)^2}$$

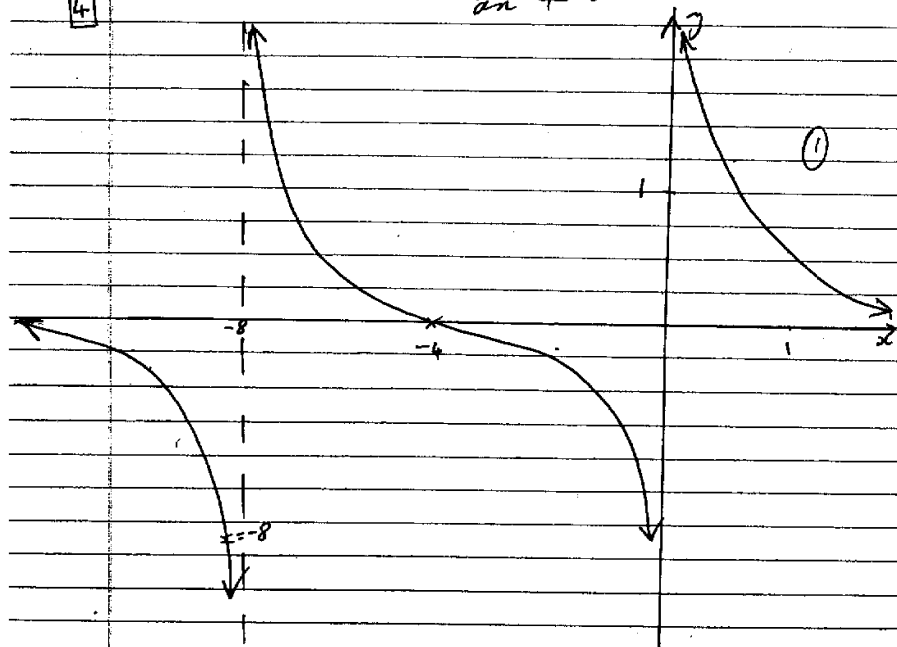
$$= 0 \text{ when } x^2+8x-2x^2-16x-32=0$$

$$x^2+8x+32=0$$

$$\text{but } (x+4)^2 + 16 \neq 0$$

$\therefore \frac{dy}{dx} \neq 0$ so no S.P.'s.

4



0

$$(d) \quad A = \int_1^2 \frac{x+4}{x^2+8x} dx$$

2

$$= \frac{1}{2} \ln(x^2+8x) \Big|_1^2$$

1

$$= \frac{1}{2} (\ln 20 - \ln 9)$$

$$= \frac{1}{2} \ln \frac{20}{9}$$

1

12

Q6 (a) (i) $f(x) = 3x - 4x^3$
 $f'(x) = 3 - 12x^2$
 $f'(a) = 3 - 12a^2$ and $f(a) = 3a - 4a^3$ (1)

(2) $\therefore$ eqn is: $y - (3a - 4a^3) = (3 - 12a^2)(x - a)$
 $y = (3 - 12a^2)x + 3a - 4a^3 - 3a + 12a^3$
 $y = (3 - 12a^2)x + 8a^3$ (1)

(iii) From (1,0):

$0 = (3 - 12a^2)1 + 8a^3$
 How many 'a' values satisfy $8a^3 - 12a^2 + 3 = 0$

Let $P(a) = 8a^3 - 12a^2 + 3$ (2)

$P'(a) = 24a^2 - 24a$
 $= 24a(a - 1)$
 $= 0$ when $a = 0$ or 1

$P''(a) = 48a - 24$
 $P''(0) = -24$
 $< 0 \therefore$ max at $(0, 3)$ (1)

$P''(1) = 24$
 $> 0 \therefore$ min at $(1, -1)$

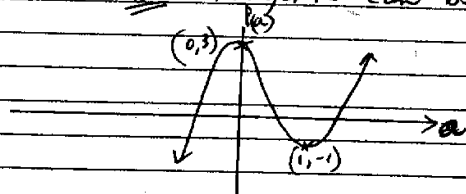
since $P(0) > 0$

and $P(1) < 0$

there must be a root between 0 & 1

$\therefore P(a) = 0$ has 3 real roots

so 3 tangents can be drawn.

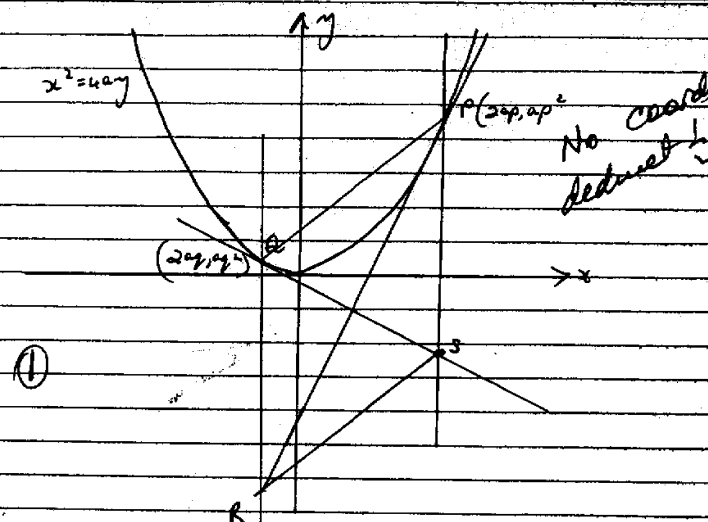


Q

(b)

Area of $\triangle PQR$ is not required.

(1)



(ii) at P: $\frac{dy}{dp} = 2ap \therefore \frac{dy}{dx} = 2ap \cdot \frac{1}{2a}$
 $\frac{dy}{dx} = p$ (1/2)

(2)

$\therefore$ Eqn PR is

$y - ap^2 = p(x - 2ap)$
 $y = px - 2ap^2 + ap^2$
 $y = px - ap^2$ (2)

(iii) PS // QR given equation of QS is $y = qx - ag^2$
 at S, $x = 2ap \therefore y = q(2ap) - ag^2$ (1)

$\therefore PS = ap^2 - (2apq - ag^2)$
 $= a(p^2 - 2pq + q^2) = a(p - q)^2$ (1)

(2)

at R, $x = 2ag \therefore y = p(2ag) - ap^2$ (1)

$\therefore QR = ag^2 - (2apq - ap^2)$
 $= a(q^2 - 2pq + p^2) = a(q - p)^2$ (2)

since $a(p - q)^2 = a(q - p)^2$
 then $PS = QR$ so PQRS is a parallelogram
 (one pair of opp sides equal & parallel).

$$\begin{aligned}
 \text{(iv) } A_{\text{area}} &= PS \times \text{perp dist from } P \text{ to } Q \\
 &= a(p-q)^2 \times |2ap-2aq| \quad (1) \\
 &= a(p-q)^2 \times 2a|p-q| \\
 &= 2a^2(p-q)^2|p-q| \\
 &= 2a^2|(p-q)^3| \quad (1) \\
 &= 2a^2|p-q|^3
 \end{aligned}$$

2

12

(iii) alternative method.

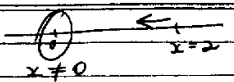
$$\begin{aligned}
 M_{PQ} &= \frac{ap^2 - aq^2}{2ap - 2aq} & M_{SR} &= \frac{2apq - ap^2 - 2apq + aq^2}{2aq - 2ap} \\
 &= \frac{p+q}{2} & &= \frac{p+q}{2}
 \end{aligned}$$

$\therefore SR \parallel PQ$
 $QR \parallel PS$

$PQRS$ is a parallelogram.

* if all that's given is $QR \parallel PS \parallel y\text{-axis}$ - $Aw \frac{1}{2}$

$Aw \frac{1}{2}$ for (2 marks)

Q7 (a) (i) $\ddot{x} = -\frac{4}{x^3}$ 

$$\frac{d}{dt}\left(\frac{1}{2}v^2\right) = -\frac{4}{x^3}$$

$$\frac{1}{2}v^2 = 2x^{-2} + C \quad (1)$$

when $t=0$, $v=0$ & $x=2$

$$\therefore 0 = \frac{1}{2} + C \Rightarrow C = -\frac{1}{2} \quad (2)$$

$$\frac{1}{2}v^2 = \frac{2}{x^2} - \frac{1}{2}$$

$$v^2 = \frac{4}{x^2} - 1 \quad (3)$$

Now at $t=0$, $v=0$ & $\ddot{x} < 0 \therefore$ moves left
But since motion is not defined for $x=0$,
particle can not ever stop

$$\text{so } v = -\sqrt{\frac{4}{x^2} - 1} \quad (1)$$

$$= -\frac{\sqrt{4-x^2}}{x} \text{ since } x > 0$$

(ii) $\frac{dx}{dt} = -\frac{\sqrt{4-x^2}}{x} \quad (1)$

$$\frac{dt}{dx} = -\frac{x}{\sqrt{4-x^2}} \quad (2)$$

3 $t = -\sqrt{4-x^2} + C \quad (1)$

when $t=0$, $x=2$ so $C=0$

$$t = \sqrt{4-x^2} \quad (2)$$

when $x=1$, $t=\sqrt{3} \quad (3)$

0

(h) (i) Let $P(n)$ be the proposition that for integral $n \geq 1$

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Test $P(1)$: LHS = $1^2 = 1$

$$\text{RHS} = \frac{1(2)(3)}{6} = 1 = \text{RHS} \quad (1)$$

$\therefore P(1)$ is true

Let k be a value of n for which $P(k)$ is true

$$\text{i.e. } 1^2 + 2^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6}$$

Now for $P(k+1)$

$$1^2 + 2^2 + \dots + k^2 + (k+1)^2 = \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \quad (1)$$

3

$$= \frac{k+1}{6} [k(2k+1) + 6(k+1)]$$

$$= \frac{k+1}{6} [2k^2 + 7k + 6] \quad (2)$$

$$= \frac{k+1}{6} (2k+3)(k+2) \quad (3)$$

$$= \frac{(k+1)(k+1+1)(2(k+1)+1)}{6}$$

as required

So provided $P(k)$ is true, it is established that $P(k+1)$ is true.

$\therefore$ Since $P(1)$ is true, then $P(2)$ is true, hence $P(3)$ is true etc

$\therefore P(n)$ is true for all integers $n \geq 1$.

(ii) $2^2 + 4^2 + 6^2 + \dots + 100^2 = 2^2 [1^2 + 2^2 + \dots + 50^2] \quad (1)$ so using $n=50$,

$$= 4 \times 50 \times 51 \times 101 = 171700 \quad (2)$$

2

1

(iii) $1^2 + 3^2 + \dots + 99^2 = (1^2 + 2^2 + \dots + 100^2) - (2^2 + 4^2 + 6^2 + \dots + 100^2)$

$$= \frac{100 \times 101 \times 201}{6} - 171700 = 166650 \quad (1)$$

12